

APOLLO 13

APOLLO 13 was almost 56 hours into its journey to the Moon and some 205,000 miles (330,000km) from Earth when, on the evening of April 13, 1970, the crew heard a sudden bang. An explosion in an oxygen tank had knocked out their command and service module's electricity supply and their remaining oxygen was leaking into space. For mission commander James Lovell, Fred Haise, and Jack Swigert, the chances of survival looked slim. Swigert must have been tempted to think himself one of the world's unluckiest men, having at the last moment taken the place of Ken Mattingly, believed to be at risk of the German measles.

The crew's one hope of returning to Earth was the undamaged condition of the lunar module, which had its own oxygen, water, power systems, and engine. By transferring to the lunar module they could survive. Mission control at Houston worked out a flight plan to bring the astronauts to safety by swinging around the Moon onto a course back to Earth. But this would take four days and the power supply on board the lunar module was only good for two.

Mission controllers and the crew working together found ways of closing down equipment



ORIGINAL CREW

The original Apollo 13 crew were, left to right, Jim Lovell, Ken Mattingly, and Fred Haise. Jack Swigert replaced Mattingly two days before the launch.

to save electricity and, in a feat of improvisation, rigged up a system for extracting the carbon dioxide the crew were exhaling from the module's unventilated atmosphere. The small cabin soon became cold and damp, temperatures falling close to freezing. Unable to sleep, with no hot food and short of water, the three astronauts

clung on. Meanwhile, on Earth their progress was followed by hundreds of millions – prayers were said, and hushed groups clustered in front of televisions in store windows.

Early on the afternoon of April 17, Apollo 13 splashed down in the Pacific just 4 miles (6 km) from its recovery ship. Shown live on television, this event may have been watched by more people than the first Moon landing. *Newsweek* described it as “the most amazing rescue operation of all time,” and few would disagree.

MISSION CONTROL

The Houston Manned Spacecraft Center acted as the mission control center throughout the Apollo program. During the Apollo 13 mission it was a scene of great tension as the drama unfolded. Instructions from controllers helped find the solutions the astronauts needed for survival.



staggering display of technological prowess, with Pete Conrad and Alan Bean landing within a few hundred yards of their target, the remains of the Surveyor 3 spacecraft. But their mission was marred by the malfunction of the camera that was to broadcast live pictures of their moonwalk. For most of the world, without the television pictures it was hard to believe or even imagine.

Ironically, it took a failed mission to raise public interest back to fever pitch. The only manned Moon shot that did not reach its goal, Apollo 13 became a drama of survival against the odds that touched the human heart in a way no perfectly executed feat of technology could (see panel, left).

The final missions

Despite possessing the hardware for more Moon landings, NASA was forced by ever-tightening budgetary constraints to limit the Apollo program to four more shots after the “successful failure” of Apollo 13. The emphasis was on scientific investigation, although only one scientist went to the Moon, geologist Harrison Schmitt, who accompanied Gene Cernan in the lunar module on the Apollo 17 mission.

Improvements in the techniques of lunar exploration continued to be made. When Apollo 14 went to the Moon in February 1971, Mercury 7 veteran Alan Shepard and his lunar module pilot Ed Mitchell were provided with the first lunar vehicle, a two-wheeled cart – rather pompously designated as a “modularized equipment transporter” – to carry equipment and samples. Apollo 15, in August 1971, brought the debut of the battery-powered Lunar Roving Vehicle, which extended the area of exploration on the Moon to a radius of 6 miles (10km) around the landing site.

Also equipped with improved pressurized suits, the crews of the last three Apollo missions were able to get through a great deal of work on the Moon. The total moon-walk time rose on each mission, reaching over 22 hours on Apollo 17. This was far from easy for the astronauts, operating in Moon gravity, which is one-sixth that of the Earth. Harrison Schmitt described how the simple operation of climbing into the Lunar Rover became a complex piece of acrobatics, involving a two-footed jump with a sideways twist, after which you “wait until you slowly settle into the seat, ideally in the correct one.” It was no small feat to collect, as Schmitt and Cernan did, about 250lb (115kg) of rock samples when simply tipping a scoop of material into a bag required total concentration.

Apollo 17 was the last manned Moon mission. After three days on the Moon's surface, Schmitt and Cernan departed on December 14, 1972. They left behind a plaque that drew a line under

STRICKEN SERVICE MODULE

The Apollo 13 service module floats dead in space, with a panel blown off its side by an explosion, exposing battery fuel cells, tanks, and other components. This was the view of the service module that the astronauts had as they prepared to reenter the Earth's atmosphere.

